

CHAPTER 2

LAWS OF PERFECT GAS

2.1 Gas Laws

Thermodynamic analysis relies largely upon the gas laws, which are known as Boyle's law, Charles's law, Gay-Lussac's law and Avogadro's law. Gases possess three important properties, pressure, volume and temperature. The relationship that exist between these properties are the basic for the gas laws.

2.1.1 Boyle's law

Boyle's law states that the pressure exerted by a gas is inversely proportional to the volume occupied by it at a constant mass and temperature, $P \propto \frac{1}{V}$. Where P is the pressure exerted by the gas and V is the volume occupied by it. This proportionality can be converted into an equation by adding a constant, k.

$P = \frac{K}{V} \Rightarrow PV = K$, means that the product of the initial pressure and the initial volume of a gas is equal to the product of final pressure and final volume. This law can be expressed mathematically as follows:

$$P_1V_1 = P_2V_2 \quad 1$$

Where,

- P_1 is the initial pressure exerted by the gas
- V_1 is the initial volume occupied by the gas
- P_2 is the final pressure exerted by the gas
- V_2 is the final volume occupied by the gas

Example 1

A fixed amount of gas occupies a volume of 1L and exerts a pressure of 400 kPa on the walls of its container. What would be the pressure exerted by the gas if it is

completely transferred into a new container having a volume of 3 liters (assuming the temperature and quantity of gas remain constant)?

Solution,

$$P_1 V_1 = P_2 V_2 \Rightarrow P_2 = \frac{P_1 V_1}{V_2} = 133.33 \text{ kPa}$$

Therefore, the gas exerts a pressure of 133.33 kPa on the walls of the 3-liter container.

Exercise 1

A gas exerts a pressure of 3 kPa on the walls of container 1. When container 1 is emptied into a 10-liter container, the pressure exerted by the gas increases to 6 kPa. Find the volume of container 1. Assume that the temperature and quantity of the gas remain constant. (The volume of container 1 is 20 L).

2.1.2 Charle's law

Charle's law states that if any gas is heated at constant pressure, its volume changes directly to its absolute temperature, $V \propto T$. Or $\frac{V}{T} = \text{constant}$, so long as pressure is constant. If the gas changes its volume from V_1 to V_2 and absolute temperature from T_1 to T_2 without any change of pressure, then

$$\frac{V_1}{T_1} = \frac{V_2}{T_2} \quad 2$$

Example 1

Find the initial volume of a gas at 300 K, if the ultimate volume is 6 L at 200 K

Solution: Given, $V_2=6$ L, $T_1= 300$ K, $T_2= 200$ K, Using Charles Law

$$V_1 = T_1 \frac{V_2}{T_2} = 300 \frac{6}{200} = 9. \text{ The initial volume of a gas at 300 K is 9 liters.}$$

Exercise 1

A gas occupies a volume of 300cm³ at 0°C (degree celsius) and 1bar What volume (in liters) will it occupy at 80°C (degree celsius) and 1bar? (ans. 0.232ltr).

2.1.3 Gay-Lussac law

Gay-Lussac's Law states that the pressure of a given mass of gas varies directly with the absolute temperature of the gas when the volume is kept constant, $P \propto T$. Or $\frac{P}{T} = \text{constant}(K)$, so long as the volume is constant. Where, P is the pressure exerted by the gas, T is the absolute temperature of the gas and K is a constant

If the gas changes its pressure from P_1 to P_2 and absolute temperature from T_1 to T_2 without any change of volume, then the formula can be expressed as follows

$$\frac{P_1}{T_1} = \frac{P_2}{T_2} \quad 3$$

Where:

- T_1 is the initial temperature
- T_2 is the final temperature

Example 1

The pressure of a gas in a cylinder when it is heated to a temperature of 250K is 1.5 atm. What was the initial temperature of the gas if its initial pressure was 1 atm?

Solution

$$T_1 = P_1 \frac{T_2}{P_2} = 1 \frac{250}{1.5} = 166.66K$$

Exercise 1

At a temperature of 300 K, the pressure of the gas in a deodorant can is 3 atm. Calculate the pressure of the gas when it is heated to 900 K. ($P_2 = 9 \text{ atm}$).

Note: Gay-Lussac's Law is very similar to Charles's Law, with the only difference being the type of container.

2.1.4 Avogadro's law

Avogadro's gas law states the volume of a gas is proportional to the number of moles of the gas at a given temperature and pressure, $V \propto n$. Or $\frac{V}{n} = \text{constant}(K)$, so long as the temperature and pressure are constant.

If a gas changes its volume from V_1 to V_2 and the number of moles from n_1 to n_2 without any change of temperature and pressure, then the formula can be expressed as follows

$$\frac{V_1}{n_1} = \frac{V_2}{n_2} \quad 4$$

- n_1 is the initial number of moles
- n_2 is the final number of moles

2.2 Ideal Gas

Ideal gas, also called perfect gas is defined as a gas having no forces of intermolecular attraction. The gases which follow the gas laws at all ranges of pressures and temperatures are considered as “ideal gases”. However, ‘real gases’ follow these laws at low pressures or high temperatures or both. This is because the forces of attraction between molecules tend to be very small at reduced pressures and elevated temperatures.

An ideal gas obeys the law $PV = RT$. The specific heat capacities are not constant but are functions of temperature. A perfect gas obeys the law $Pv = RT$ and has constant specific heat capacities [1].

2.2.1 The Equation of State for a Perfect Gas

Any equation that relates, between the independent properties, such as pressure, specific volume and temperature for a pure substance is known as the

‘**equation of state**’. There are several equations of state, some simple and others are complex. The simplest and best-known equation of the state for substances in the gas phase is the **ideal-gas equation of state**. Thus, the combination of the gas laws, gives the ideal-gas equation of state below:

$$PV = mRT \quad 5$$

$$Pv = RT \quad 6$$

$$\frac{Pv}{T} = R \quad 7$$

If there are n moles present then the ideal gas equation may be written as

$$PV = nRT \quad 8$$

Where P , V , and T are the pressure, volume and temperature of gas having mass m , v =specific volume= m/V , n is number of moles and **R** is a constant for the gas known as **gas constant**.

The gas constant R is different for each gas and is determined from

$$R = \frac{R_u}{M} \quad [\text{KJ/kg.K or kPa.m}^3/\text{kg.K}] \quad 9$$

Universal gas constant (R_u) and molar mass (M)

R_u is the universal constant and its value is the same for all substances as **8.3147** kJ/kmol.K or 8.31447 kPa.m³/mol.K and M is the molar mass (molecular weight) of the gas.

For the two states of the gas, eqn. (7) can be written as

$$\frac{P_1 V_1}{T_1} = \frac{P_2 V_2}{T_2} \quad 10$$

Example 1

A vessel of 0.03 m³ capacity contains gas at 3.5 bar pressure and 35°C temperature. Determine the mass of the gas in the vessel. If the pressure of this gas is increased to 10.5 bar while the volume remains constant, what will be the temperature of the gas? For the gas take $R = 290$ J/kg K. (ans. 0.12kg, 650°C).

Solution

Using equation of state,

$$PV = mRT$$

$$m = \frac{PV}{RT} = \frac{3.5 \times 10^5 \times 0.03}{290 \times 308} = 0.118 \text{ kg}$$

$$T_2 = T_1 \frac{P_2}{P_1} = 308 \frac{10.5}{3.5} = 924K \quad \text{Or } 651^\circ\text{C, where } V_1=V_2$$

2.2.2 Specific heats

Quantity of Heat, $Q = \text{mass} \times \text{specific heat capacity} \times \text{temp. diff.}$ 11

$$Q = mCdT$$

This heat exchange may take place, either at constant volume, $Q = C_v(T_2 - T_1)$ Or at constant pressure, $Q = mC_p(T_2 - T_1)$,

Where, C_v specific heat at constant volume (kJ/kg K) and C_p specific heat at constant pressure (kJ/kg K),

Note that for a perfect gas $C_p = C_v + R$ and $\gamma = \frac{C_p}{C_v}$

Thus, the amount of energy needed to raise the temperature of a unit mass of a substance by one degree is called the specific heat at constant volume C_v for a constant-volume process or constant pressure C_p for a constant- pressure process. They are defined as [4]

i. Specific heat at constant volume, $C_v = \frac{du}{dT}$

ii. Specific heat at constant pressure, $C_p = \frac{dh}{dT}$

2.2.3 Specific Heat at Constant Volume C_v

Consider a closed system of unit mass, the first law of thermodynamics applied to this system is:-

$$q - w = du \quad 12$$

If the volume is constant. Then, $w = 0$, the equation becomes

$$q = du \quad 13$$

But

$$q = du = C_v dT$$

$$C_v = \frac{du}{dT}$$

This is **known as Joule's Law of internal energy** which states that "the internal energy of a perfect gas depends upon its temperature only".

2.2.4 Specific Heat at Constant Pressure C_p

Consider a constant pressure non-flow process, the first law:-

$$q - w = du$$

For constant pressure

$$w = P(V_2 - V_1)$$

Consider a constant pressure non-flow process, the first law:-

$$q - w = du$$

For a constant pressure process

$$w = P(V_2 - V_1) = P_2V_2 - P_1V_1$$

Hence

$$q - P_2V_2 - P_1V_1 = u_2 - u_1$$

OR

$$q = u_2 + P_2V_2 - (u_1 + P_1V_1)$$

$$q = h_2 - h_1$$

but

$$q = c_p dT$$

Hence

$$h_2 - h_1 = c_p(T - T_1)$$

$$C_p = \frac{dh}{dT}$$

2.2.5 Relationship between Specific Heats

Since

$$H = U + PV$$

$$dH = dU + d(PV)$$

$$mc_p dT = mc_v dT + mRdT$$

$$c_p = c_v + R$$

The ratio $\frac{c_p}{c_v} = \gamma$ is called the adiabatic index [1].

$$C_p = C_v + R \quad 14$$

$$\gamma = \frac{C_p}{C_v} \quad 15$$

$$C_v = \frac{R}{\gamma - 1} \quad 16$$

$$C_p = \frac{\gamma R}{\gamma - 1} \quad 17$$

Example 1

A closed rigid container has a volume of 1 m^3 and holds air at 345 kPa and 20°C .

Heat is added until the temperature is 327°C . Determine the change in Internal Energy ($C_v = 741 \text{ J/kgK}$ and $R = 287 \text{ J/kgK}$).

Given, $V = 1 \text{ m}^3$, $P = 345 \text{ kPa}$, $T_1 = 20 + 273 = 293 \text{ K}$ and $T_2 = 327 + 273 = 600 \text{ K}$, $C_v = 741 \text{ J/kgK}$ and $R = 287 \text{ J/kgK}$

Solution

A rigid container means the process took place at constant volume, therefore this formula is to be used

$$Q = mC_v(T_2 - T_1)$$

Mass is not given it needs to be calculated from the equation of state

$$m = \frac{PV}{RT} = m = \frac{1 \times 345 \times 10^3}{287 \times 293} = 4.10 \text{ kg}$$

$$Q = mC_v(T_2 - T_1) = 4.10 \times 0.741(600 - 293) = 933 \text{ kJ}$$

Example 2

A perfect gas has a molecular weight of 28 and a value of $\gamma=1.11$. Calculate the heat rejected per kg of gases.

- When the gas is contained in a rigid vessel of 5 bar and 350°C. It is then cooled until the pressure falls to 1.5 bar
- When the gas enters a pipeline at 300 °C and flows steadily to the end of the pipe where the temperature is 50 °C. Neglect changes in velocity of the gas in the pipeline.

Solution

Given

- i) $R=28$, $\gamma=1.11$, $\frac{Q}{m} = C_v(T_2 - T_1)$, rigid container means constant volume,

$$T_1=350+273\text{K}=573\text{K}, P_1=5\text{Pa}, P_2=1.5\text{bar. find } C_v$$

$$C_v = \frac{R}{\gamma-1} = \frac{0.2969}{1.11-1} = 2.699 \text{ kJ/kgK},$$

$$R=R_u/M=8.3144/28=0.2969 \text{ kJ/kgK}$$

Constant volume law,

$$T_2 = T_1 \frac{P_2}{P_1} = 573 \frac{1.5}{5} = 186.9 \text{ K}$$

$$\text{Therefore, } \frac{Q}{m} = q = C_v(T_2 - T_1) = 2.699(186.9 - 573) = -1,177 \text{ J/kgK}$$

- i. $T_1=300+273=573\text{K}$, steadily mean constant pressure, C_p is to be used and

$$T_2=50+273=323\text{K. find } C_p=?=C_v + R=2.699+0.2969=2.997$$

$$q = C_p(T_2 - T_1) = 2.997(323 - 573) = -749 \text{ J/kgK}$$

2.3 Assignment 2

1. List 4 gas laws and state any 3 with their formulae
2. What is an ideal gas? Write ideal gas equation of state and define all its parameters and their units and define the universal gas constant.
3. 6. State Joule's law and derive the relationship between the two specific heats characteristic gas constant and gamma for a perfect gas.
4. Proof that $C_p = \frac{\gamma R}{\gamma - 1}$
5. An unknown gas has a mass of 1.5 kg contained in a bottle of volume 1.17 m^3 while at a temperature of 300 K, and a pressure of 200 kPa. Determine the ideal gas constant and deduce the molar mass of the gas? (ans. $R=520 \text{ J/kgK}$ and $M=15.99$).
6. Determine the pressure of water vapour at 300°C , if its specific volume is $0.2579 \text{ m}^3/\text{kg}$, and gas constant is 461.5 J/kgK . (Ans. 10.3 bar).
7. Calculate the density of Ethane at 171 bar and 458K. Assume R for ethane is 319.3 J/kgK . (ans. 117 kg / m^3).[2].
8. What are semi-perfect or permanent gases?